

Analysis PhD Exam, January 1993

Be sure to present all work in a neat and logical fashion in order to receive credit.

1. State and prove the Radon–Nikodym theorem.
2. Let (S, Σ, μ) be a positive finite measure space. Prove that $(L^1(S, \Sigma, \mu))^* = L^\infty(S, \Sigma, \mu)$.
3. Give the construction of the product measure $\mu \times \nu$ on $\mathcal{S} \otimes \mathcal{W}$, where $(X, \mathcal{S}, \mu)$ and $(Y, \mathcal{W}, \nu)$ are finite non negative measure spaces. (Be sure to prove that $\mu \times \nu$ is countably additive.)
4. Let (S, Σ, μ) be a positive measure space with $\mu(S) = 1$. Let $\mathcal{A}$ and $\mathcal{B}$ be a sub σ -algebras of Σ . Suppose $\mu(A \cap B) = \mu(A)\mu(B)$ for every $A \in \mathcal{A}$ and $B \in \mathcal{B}$. Show that $\int fg d\mu = (\int f d\mu)(\int g d\mu)$, for $f \geq 0$, $g \geq 0$, where f is $\mathcal{A}$ -measurable and g is $\mathcal{B}$ -measurable.
5. Let (S, Σ) be a measurable space, and suppose μ and ν are positive finite measures on Σ . Suppose $\mu \ll \nu$ and $\nu \ll \mu$. Show that

$$\frac{d\mu}{d\nu} = \frac{1}{\frac{d\nu}{d\mu}} \quad \text{a.e. } \nu.$$

6. Let (S_∞, Σ, μ) be a finite measure space. Let $(f_n)_{n=1}^\infty$ be a sequence of integrable function such that

$$\sum_{n=1}^\infty \int_S |f_n| d\mu < \infty.$$

Does it follow that $\sum f_n$ converges pointwise on S a.e. μ ? Show all details.

7. Let f be a Lebesgue integrable function on $\mathbb{R}$. Let $\epsilon > 0$. Is it possible to find a bounded interval I such that whenever E is a measurable set such that $E \cap I = \emptyset$, then

$$\left| \int_E f dm \right| < \epsilon?$$

Show all work.

8. Let f be Lebesgue integrable on $[0, 1]$. Suppose that $\int_0^x f dm = 0$, for every $x \in [0, 1]$. What can you conclude about f ? Prove your answer?